

Title of the Paper	Author(s) & Year	Conference	Summary (Abstract)	Similarities (with our function)	Uniqueness (our function)
Seaweed Production, Trade and Marketing Status in Sri Lanka	Jayasuriya, P.M.A. (2022)	Local Publication (Possibly FAO-related)	Reviews historical and current trends in seaweed production, focusing on Sargassum and Gracilaria spp. for domestic use and export. Discusses harvesting methods, export trends, and socio-economic impacts on fishing communities.	Shares focus on Sri Lankan seaweed species and economic relevance. Mentions species-level data and use cases.	Does not use AI or image recognition. Your function provides digital classification and real-time identification with edibility/use feedback.
Distribution and Diversity of Seaweed Species in South Coastal Waters in Sri Lanka	Premarathna, A.D. et al. (2020)	Journal of Oceanography and Marine Research	Explores species diversity and ecological patterns of seaweed (e.g., Sargassum, Ulva) along the south coast using ecological survey methods. Provides species-specific habitat data and water parameter correlations.	Addresses seaweed species identification and classification. Uses species traits and habitats similar to traits stored in your system.	Uses CNN and image uploads for recognition. Your tool is user-interactive and works without physical sampling or field visits.
Chemistry and Biological Activity Studies of Green Algae Ulva lactuca L.	Haniffa, H.M. (2021)	Journal of Science-FAS-SEUSL	Evaluates phytochemical compounds and cytotoxicity in Ulva lactuca extracts. Identifies nutritional and anti-cancer properties and provides detailed chemical isolation methods.	Uses Ulva lactuca as a key species; mentions biological importance and identification.	Not chemical or lab-based. Uses AI vision and offers multi-species classification with instant web interface.
Seaweed Aquaculture in Sri Lanka - Challenges and Policy Recommendations	Gamage, N.D. & Nagarajah, S. (2021)	ResearchGate Preprint / Conference Paper	Presents challenges in Sri Lanka's seaweed farming sector including lack of technical knowledge, market support, and cultivation methods. Suggests policy strategies for sustainable growth.	Shares goal of improving seaweed farming and support systems.	Your solution provides hands-on species-level tech support using AI for real-time recognition, which this paper doesn't address.
The Status of Culture and Utilization of Seaweeds in Sri Lanka	Jayasuriya, P.M.A. (1989)	FAO Regional Report	Covers the state of seaweed harvesting, export, and utilization in Sri Lanka from 1950s–1980s. Details economic uses and regional cultivation attempts of Gracilaria and Sargassum.	Focuses on Sri Lankan seaweed types and their uses (e.g., food, pharma). Discusses species and local practices.	No technological or AI implementation. Your function modernizes the process through CNN image classification and interactive database.

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Development of Green Seaweed (Ulva fasciata) Incorporated String Hoppers as a Functional Food	Fernando, M. & Wijesekara, I. (2021)	IEEE FITI 2021	Investigates the use of Ulva fasciata in functional food, incorporating it into string hoppers to enhance nutrition. Reports on phytochemical, proximate, and sensory analysis with 2.5% incorporation found optimal.	Uses Ulva fasciata, involves morphological and chemical identification of seaweed species.	Your function offers automated species classification via CNN and image upload, whereas this work is lab-based nutritional enhancement.
Investigating the Possibilities to Produce Alginate Filaments with Locally Available Seaweeds	Dharmarathne, A. et al. (2020)	MERCon 2020	Explores extraction of alginate from Sargassum and creation of textile filaments. Studies blends of alginate with clay, silicate, and graphene oxide for tensile and moisture properties.	Focuses on brown seaweed like Sargassum and includes chemical extraction and utility data.	Does not use AI or image classification. Your system classifies species and gives use-case info to non-experts through an app interface.
Comparative Study on Three Selected Culture Methods for Padina antillarum	Bambaranda, B.V.A.S.M. et al. (2023)	Journal of Technology and Value Addition	Compares three seaweed cultivation methods (raft, longline, bag) for brown seaweed P. antillarum on Sri Lanka's Mannar coast. Measures growth, AGR, and WGR.	Focuses on specific seaweed species and growth behavior, useful in defining characteristic data.	Your work automates species detection and does not rely on physical culture methods or AGR calculations.
Sri Lankan Seaweeds: Methodologies and Field Guide to the Dominant Species	Coppejans, E. et al. (2009)	ABC Taxa, Vol. 6	Comprehensive visual guide to Sri Lanka's seaweed species with taxonomic keys, collection zones, and manual ID charts.	Provides species-level trait data for Sri Lankan seaweeds; your database builds on similar visual ID logic.	Uses AI/CNN for automatic image-based classification. Also enables real-time feedback and filtering irrelevant uploads.
Composition Analysis of Selected Sri Lankan Seaweeds	Jayakody, M.M. et al. (2019)	Journal of Tropical Forestry and Environment	Performs proximate, sulfate, fiber, and mineral content analysis on Ulva fasciata, Chnoospora minima, and Porphyra sp. using AOAC methods.	Covers Ulva fasciata and provides nutritional/chemical traits that may align with species database design.	Not automated or vision-based. Your model operates without needing chemical lab equipment, directly from photos.

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Automated Detection and Classification of Medicinal Plant Leaf Diseases using CNNs	Sonkar, A. et al. (2025)	IEEE ICAIHC 2025	This paper proposes a CNN-based system for identifying fungal infections in medicinal plant leaves using image processing. The system achieved 98.87% accuracy and uses a soft-computing CNN model with ReLU and SoftMax activation functions.	Uses CNN for disease classification from leaf images. Fully automated diagnosis with real-time accuracy.	Your system targets seaweed, not land plants, and includes edibility, usage, safety info, and rejection of irrelevant uploads.
Automated Detection of Tomato Leaf Disease Using Convolutional Neural Networks	Santhiya, S. et al. (2024)	IEEE ICTBIG 2024	Compares GoogLeNet, AlexNet, and a custom CNN to detect tomato leaf disease. GoogLeNet achieved highest accuracy at 92%. Focus is on deep learning architectures and image-based detection.	Applies CNN architectures to detect plant disease via image analysis.	Your function uses a single CNN tailored for marine species (seaweed) and integrates backend-edibility data, not just disease detection.
Role of Convolutional Neural Networks in Plant Leaf Disease Detection	Singh, M.N. et al. (2024)	IEEE ICPCSN 2024	A review of CNN applications in identifying plant leaf diseases using image recognition. Evaluates various CNN models and identifies gaps in generalization and real-world deployment.	Both explore CNNs for identifying plant types/diseases via images.	Your function is species-level classification with non-disease focus and web app integration for user feedback and education.
Optimal Feature Selection and Classification Using CNN-Based Plant Disease Prediction	Lakineni, P.K. et al. (2023)	IEEE ICICACS 2023	Combines Logistic Decision Regression with CNN for improved disease classification. Achieves 95% accuracy on a 38-class plant dataset. Emphasizes preprocessing and optimized image recognition.	CNN-based classification from image datasets with accuracy benchmarking.	No use of feature engineering like LDR. Your system handles noise, invalid inputs, and returns contextual species information for farmers.
CNN-Based Model for Real-Time Disease Detection in Agriculture	Sharma, R. et al. (2023)	IEEE (from ml1.pdf)	Develops a CNN for real-time plant disease classification with 94.8% accuracy. Demonstrates field deployability and integration with sensors for live monitoring.	Focuses on real-time CNN inference for identifying plant condition from image data.	Your focus is not just classification but education and species-level identification in aquatic plants with rejection handling for invalid images.

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Waste Detection in Underwater Using Deep Vision Transformers	Gupta et al., 2025	ICDSAAI 2025, IEEE	Uses Vision Transformers for real-time underwater waste detection. Overcomes CNN & YOLO limitations using self-attention mechanisms.	Both systems use deep learning and image recognition in marine environments. Focus on classification accuracy in complex underwater settings.	Our function detects seaweed specifically, not waste. It provides edibility, uses, warnings, and rejects non-seaweed item unique to seaweed use cases.
An Efficient Model for Plant Disease Detection in Agriculture Using Deep Learning Approaches	Puneeth N. Thotad et al., 2023	IEEE GCAT 2023	CNN-based plant disease classifier trained on 54,306 images. Web application detects leaf disease with 95% accuracy.	Similar use of CNNs, leaf-based image classification, and full-stack web application.	Ours focuses on marine organisms, includes MongoDB integration, and detects invalid objects (non-seaweed), which this system lacks.
Medicinal Plant Identification Using CNN and Pytorch Lightning	Thilagam T. et al., 2024	ICSTSDG 2024, IEEE	CNN model for classifying medicinal plant species with 90.33% accuracy. Uses PyTorch Lightning for modular design.	Like our system, it uses CNN and image features for plant recognition. Also addresses dataset limitations.	Unique focus on seaweed; ours adds metadata (edibility, uses), user upload interface, and backend validations not covered here.